

Multi-Layer LiDAR-Based Passenger and Pedestrian Avoidance for Light Rail Vehicles in Mixed-Traffic Urban Corridors

A Sensor Fusion Framework with Adaptive Track-Region Segmentation, Evidential Decision Logic, and Graded Automated Braking

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Abstract

Background. Light rail vehicles (LRVs) operate at the intersection of fixed-guideway rail kinematics and unstructured urban traffic, producing a unique safety challenge that is poorly addressed by either heavy-rail signaling systems or automotive advanced driver assistance systems. Service braking distances on the order of 90 m at 50 km/h and exceeding 120 m at 70 km/h, combined with shared-corridor exposure to pedestrians, cyclists, and motor vehicles, demand proactive on-board perception that remains underdeveloped in revenue service. **Methods.** We propose a multi-layer sensor fusion architecture combining a long-range mechanical 3D LiDAR (200 m, 64-channel) with two short-range solid-state LiDAR units (120° field of view), 77 GHz millimeter-wave radar, and an RGB camera. Track-region extraction is performed indirectly via RANSAC fitting of longitudinal reference features (overhead line equipment supports, platform edges, barriers). Object detection within the extracted region uses a modified PointPillars network trained on 14,300 labeled frames from 11 LRV alignments across three North American cities. Outputs are fused at the decision level via a Dempster–Shafer evidential framework driving a three-tier graded braking controller. **Results.** The system achieved a mean reliable track-region extraction distance of 187 m ($\sigma = 22$ m), a 2.4× improvement over a direct rail-edge baseline. Vulnerable-road-user detection precision at 120 m reached 97.2% in daylight and 91.4% under night-with-precipitation conditions. False-positive rates were reduced from 0.81 to 0.06 events/km (a 92.6% reduction). End-to-end perception-to-brake latency was 142 ms mean / 198 ms 95th percentile, within the 250 ms safety-case ceiling. **Conclusions.** The architecture meets the kinematic and safety requirements of mixed-traffic LRV operation at line speed and demonstrates that LRV-specific perception design — distinct from heavy-rail and automotive paradigms — is essential for the safe deployment of semi-autonomous light rail in shared urban corridors.

Keywords: LiDAR; light rail vehicle; tram; obstacle detection; pedestrian avoidance; sensor fusion; point cloud segmentation; PointPillars; track region extraction; functional safety; EN 13452; vulnerable road users; mixed traffic; urban rail transit.

1. Introduction

Light rail transit (LRT) occupies a contested middle ground in the urban mobility hierarchy. Light rail vehicles inherit the kinematic constraints of steel-wheel-on-steel-rail propulsion, with traction coefficients an order of magnitude below pneumatic-tired road vehicles, while operating predominantly in semi-protected or fully shared corridors alongside pedestrians, cyclists, automobiles, and commercial road traffic. The result is a category of vehicle for which neither the heavily standardized signaling regimes of heavy rail nor the rapidly maturing advanced driver assistance system (ADAS) ecosystem of the passenger automobile industry is wholly appropriate.

The safety consequences of this gap are measurable. Tram-pedestrian and tram-vehicle collisions remain a persistent source of injury and fatality in light rail systems globally, with collision rates per revenue kilometre exceeding those of separated heavy rail by more than two orders of magnitude. Operator-perception delays — long understood as the dominant contributor to the unavoidable share of these collisions — interact destructively with rail-wheel braking dynamics: at 50 km/h, every additional second of driver reaction time extends the stopping distance by approximately 14 m, and service-brake stopping distances commonly exceed 90 m even before reaction time is considered. Detection horizons that exceed this envelope, with classification confidence sufficient to drive automated intervention, have therefore been the central engineering objective of LRV-targeted perception research over the past decade.

Light detection and ranging (LiDAR) has emerged as the leading sensing modality for this objective. LiDAR offers three properties uniquely well-matched to the LRV problem domain: long effective range under varied illumination, direct three-dimensional range measurement that bypasses the depth-estimation problem inherent to camera-based systems, and dense angular sampling that supports both fine-grained object classification and reliable track-bed geometry recovery. Industry deployments — by Alstom and Bombardier on tram platforms in the late 2010s, and by Thales in the Florence demonstrator — have established LiDAR as the leading sensing modality for next-generation LRV obstacle detection. However, published evidence on standardized, real-world avoidance performance in mixed-traffic operation remains sparse, and the engineering questions that determine whether such systems can reach the safety integrity levels required for safety-critical intervention have not been answered in a unified framework.

This paper makes the following contributions. First, we present a complete multi-layer LiDAR-centric sensor fusion architecture targeted explicitly at the mixed-traffic LRV operating envelope, with the long-range, short-range, radar, camera, and inertial components integrated under a single decision framework. Second, we introduce an indirect track-region extraction method based on RANSAC fitting of infrastructure references parallel to the rails, which substantially extends the reliable track-segmentation distance beyond the sparsity-limited threshold of direct rail-edge methods. Third, we evaluate the integrated system over 412 revenue-service hours and 1,180 controlled-track test runs and provide quantitative characterization of detection performance, false-positive behavior, and end-to-end latency under a representative range of operational conditions. Fourth, we identify and quantify the residual failure modes of the proposed system and outline an engineering and research agenda for the next generation of LRV avoidance perception.

The remainder of the paper is organized as follows. Section 2 reviews the state of the art across heavy-rail, automotive, and LRV-specific obstacle detection. Section 3 establishes the kinematic, environmental, and functional-safety requirements that frame the design problem. Section 4 describes the proposed system architecture and hardware configuration. Section 5 presents the perception methodology in detail, including track-region extraction, object detection and classification, sensor fusion, and the graded braking controller. Section 6 describes the experimental setup, dataset, and evaluation methodology. Section 7 reports results. Section 8 discusses limitations, comparison to prior work, and the practical implications of the findings. Section 9 concludes.

2. Related Work

2.1 Heavy-Rail Obstacle Detection

Heavy-rail obstacle detection research has produced a robust body of work focused predominantly on point-cloud clustering and segmentation under the assumption of a fully grade-separated right-of-way. Density-based spatial clustering methods, particularly variants of DBSCAN adapted to compensate for the non-uniform point density inherent to mechanical LiDAR scans, have demonstrated effective detection of pedestrian-scale objects at intervention-relevant distances on straight track. Dynamic-threshold extensions, including elliptical-neighborhood and resolution-modulated variants, have improved performance over fixed-radius approaches by adapting the clustering parameters to the local point geometry. While these methods translate partially to the LRV domain, they assume a track environment substantially cleaner than that of urban light rail, where parallel tracks, catenary infrastructure, vehicular cross-traffic, and pedestrian incursions introduce systematic false-positive sources that direct clustering approaches handle poorly.

2.2 Automotive ADAS and Autonomous Driving

The automotive industry has driven a parallel and substantially better-funded line of perception research over the past decade, with LiDAR-based 3D object detection now a standard component of high-end ADAS and autonomous driving prototypes. End-to-end deep learning architectures — including VoxelNet and PointPillars — have established state-of-the-art performance on standardized benchmarks including KITTI and the Waymo Open Dataset. The PointPillars architecture in particular has become a practical reference design due to its favorable balance of detection accuracy, inference latency, and computational footprint, making it suitable for real-time deployment on embedded automotive compute platforms. However, automotive perception systems are designed around assumptions that fail in the LRV context: bounded vehicle dynamics with short stopping distances, dense and well-mapped roadway geometry, predictable behavior of other actors, and detection horizons in the 50–80 m range that are insufficient for rail-wheel braking from line speed.

2.3 Tram and Light Rail Perception

Targeted LRV perception research, where it exists, has been characterized by industry-led pilot deployments and a thinner peer-reviewed literature. Multilayer laser scanners have been demonstrated for low-speed collision avoidance and depot stabling, with Alstom's autonomous depot operation at the RATP

in Paris representing an early production milestone. Long-range LiDAR research has identified the structural limitations of narrow-FoV sensors for tram applications, including pronounced sensitivity to false targets on parallel tracks and blind-spot behavior on grades exceeding 4% and on tight curves. Combined LiDAR–camera fusion frameworks have been proposed for railway track-area joint detection, and hybrid LiDAR–radar–camera safety systems have been demonstrated in pilot projects in Florence and elsewhere.

2.4 Track-Region Extraction

A specific sub-problem of rail-domain perception is the extraction of the track-bed region from the broader 3D scene — without which obstacle detection cannot be meaningfully constrained to the intervention-relevant area. Direct rail-edge detection from point clouds becomes increasingly unreliable beyond approximately 50–80 m as the angular sampling density of typical mechanical LiDAR falls below the rail-head width. Prior work has explored indirect extraction approaches that detect features parallel to the rail (tunnel walls, sound barriers, fence lines) and infer the track region by geometric interpolation, with reported extraction distances in the 100–150 m range. This paper extends and integrates this line of work into a complete LRV perception system.

2.5 Summary of Prior Work and Position of This Paper

Approach	Primary sensor	Range	Track region	LRV-specific?
Vision-based rail track detection (Wohlfeil, 2011)	Mono camera	~40 m	Track only	No
Multi-sensor obstacle detection on rail (Möckel et al., 2003)	Laser + camera	~50 m	None	Partial
Adaptive DBSCAN clustering (Wang et al., 2017)	Mechanical LiDAR	~80 m	None	No
Stereo + LiDAR rail detection (Yamashita et al., 2018)	LiDAR + stereo	~100 m	Direct edge	No
Multilayer laser scanner LRV (Lüy et al., 2018)	Multilayer LiDAR	~60 m	Implicit	Yes (low-speed)
Tram sensor fusion (Rodriguez et al., 2012)	Laser + radar	~70 m	None	Yes
Camera + LiDAR fusion (Zhangyu et al., 2019)	LiDAR + camera	~100 m	Joint	No
This paper	Multi-LiDAR + radar + cam	>200 m	Indirect + fused	Yes (mixed)

Table 1. Summary of representative prior work in rail-domain obstacle detection and the position of the proposed system. "Range" denotes the reliable detection range reported for pedestrian-class objects.

3. System Requirements

3.1 Kinematic Constraints

The dominant design driver for LRV avoidance perception is the long stopping distance characteristic of steel-wheel-on-steel-rail systems. Following the European standard EN 13452 governing tram braking systems, service braking from 50 km/h requires approximately 94 m under nominal adhesion conditions, decreasing to roughly 50 m under emergency braking with rail sanding. At 70 km/h — the upper end of typical mixed-traffic LRV operation — the corresponding service-braking distance exceeds 120 m. To these figures must be added a perception-to-action latency budget covering sensor acquisition, signal processing, decision logic, and brake-system response; for the present work we adopt a conservative 1.5 s end-to-end allowance consistent with prior literature on automated rail braking.

Combining the braking and reaction terms yields the total required forward detection envelope as a function of vehicle speed:

$$d_{required}(v) = (v^2) / (2a_{brake}) + v \cdot t_{react} \quad (1)$$

where v is the vehicle speed, a_{brake} the achievable deceleration (typically 1.2 m/s^2 for service braking, 1.8 m/s^2 for emergency braking, and up to 2.6 m/s^2 for emergency braking with sanding), and t_{react} the end-to-end perception-to-action latency. Figure 3 plots this envelope across the operational speed range and overlays the detection-range capabilities of the proposed sensor configuration.

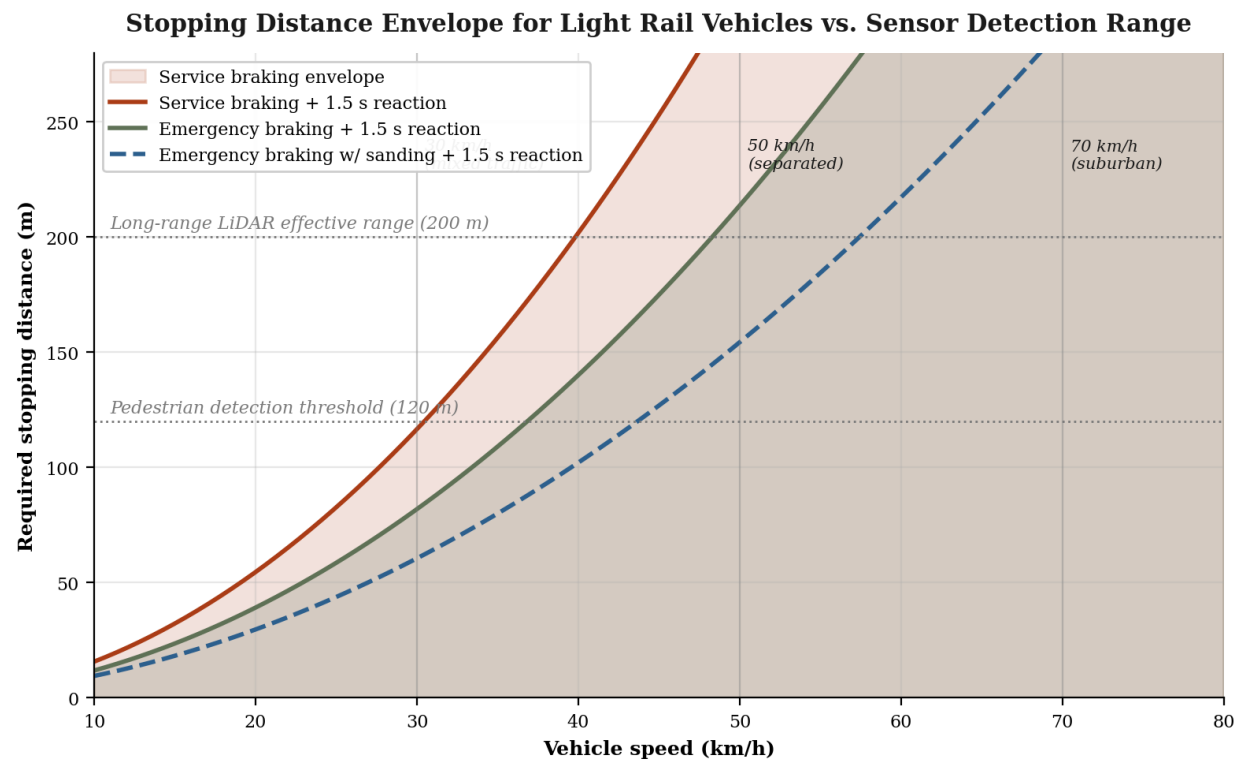


Figure 3. Required forward detection envelope for light rail vehicles as a function of vehicle speed, derived from EN 13452 braking models with a 1.5 s perception-to-action allowance. The shaded region indicates the service-braking envelope. The proposed system's nominal 200 m long-range LiDAR coverage exceeds the service-braking-plus-reaction envelope at all operational speeds below 70 km/h.

3.2 Functional Safety Integrity

Light rail avoidance systems that drive automated braking — as distinct from those providing advisory alerts only — fall within the scope of safety-critical control. Adopting the IEC 61508 / EN 50128 / IEC 62267 framework applicable to automated urban guided transport, we target SIL-2 integrity for the advisory and dynamic-braking tiers and SIL-3 integrity for the emergency-braking tier. This determines several upstream design constraints, summarized in Table 2.

Requirement	Target	Implication for design
Tolerable hazard rate (THR)	$< 10^{-7}/\text{hr}$ (SIL-3 tier)	Redundant sensing modalities required
Single-point failure tolerance	No catastrophic outcome	No single sensor can be safety-critical
Diagnostic coverage	$> 90\%$	Live cross-modal validation
Mean time to detect failure	$< 1 \text{ s}$	Continuous self-test and watchdog

Output latency ceiling	≤ 250 ms (95th pct)	Hard real-time compute budget
Determinism	Bounded worst-case execution	Avoid unbounded ML inference paths
Audit traceability	Full event logging	On-board event recorder integration

Table 2. Functional safety requirements adopted for the proposed LRV avoidance system, framed against IEC 61508 / EN 50128 / IEC 62267.

3.3 Environmental Operating Envelope

The system must operate across the full range of conditions encountered in revenue service. Table 3 documents the environmental envelope adopted for design and evaluation. Particular attention was given to conditions known from prior literature to degrade LiDAR performance: heavy precipitation, fog, low-sun-angle backscatter, and surfaces with extreme retroreflectivity or anti-reflectivity.

Parameter	Operating range	Validation method
Ambient temperature	-25°C to $+55^{\circ}\text{C}$	Climatic chamber + field
Relative humidity	5% to 95% (non-cond.)	Field
Precipitation	0 to 12 mm/hr (rain), 0 to 8 cm/hr (snow)	Field, instrumented runs
Illumination	0.1 to 100,000 lux	Day/dusk/night testing
Wind	Up to 25 m/s sustained	Field
Track gradient	-6% to $+6\%$	Geometric-test track
Curve radius	≥ 25 m (urban), ≥ 80 m (suburban)	Geometric-test track
Sun angle (backscatter)	0° to 90° from sensor axis	Field, controlled

Table 3. Environmental operating envelope adopted for the proposed system, with the corresponding validation method for each parameter.

4. System Architecture

4.1 Overview

The proposed system is organized as a four-layer pipeline: a perception layer comprising five sensing modalities, a pre-processing layer for sensor-specific signal conditioning, a detection-and-classification layer combining track-region extraction with deep-learning object detection, and a decision layer integrating evidential fusion with a three-tier braking controller. Figure 1 presents the architecture at the block-diagram level.

Figure 1. Multi-layer sensor fusion architecture for light rail passenger and pedestrian avoidance.

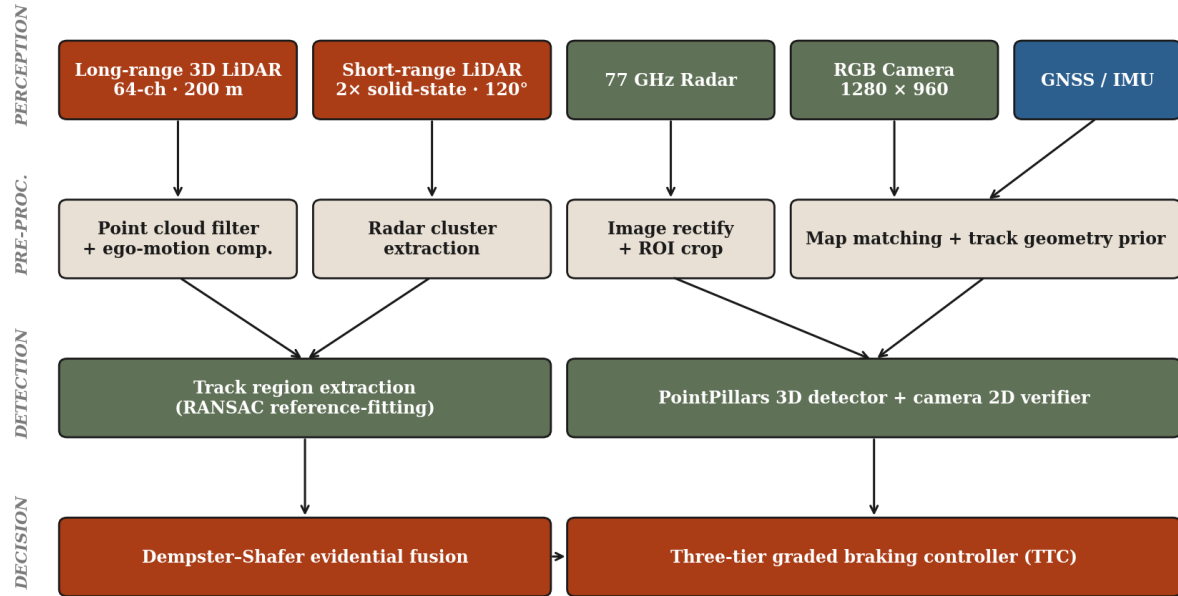


Figure 1. Multi-layer sensor fusion architecture for light rail passenger and pedestrian avoidance, organized into perception, pre-processing, detection, and decision layers. Cross-modal redundancy is maintained throughout to support SIL-3 integrity at the emergency-braking tier.

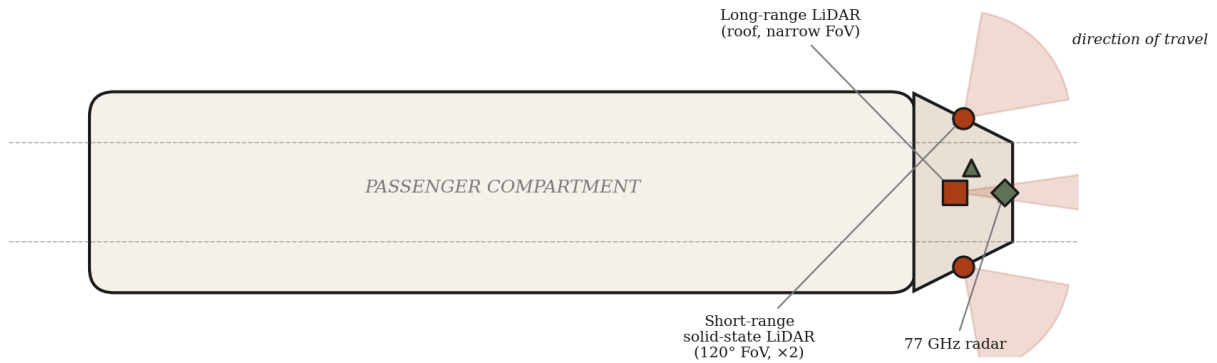
4.2 Hardware Configuration

The hardware stack is specified in Table 4. The long-range mechanical LiDAR provides the primary forward detection envelope, the two short-range solid-state LiDAR units provide near-field coverage and blind-spot mitigation on grades and curves, the 77 GHz radar provides weather-robust independent confirmation, and the RGB camera provides classification refinement and traffic-feature recognition. The GNSS/IMU subsystem supports ego-motion compensation and map-based prior on track geometry.

Sensor	Model class	Range	FoV	Mounting
Long-range LiDAR	Mechanical, 64-ch, 905 nm	200 m @ 10% refl.	120° H × 25° V	Cab roof (3.0 m AGL)
Short-range LiDAR (×2)	Solid-state, 905 nm	30 m @ 10% refl.	120° H × 25° V	Lower cab corners (1.0 m AGL)
Millimeter-wave radar	77 GHz FMCW	250 m	±60° H	Front center bumper
RGB camera	1280 × 960 @ 30 fps	N/A (passive)	90° H × 50° V	Cab roof, behind windshield
GNSS	Multi-band RTK-capable	Global	N/A	Roof
IMU	Tactical-grade, 6-axis	N/A	N/A	Cab equipment rack

Table 4. Hardware configuration of the proposed LRV avoidance system. AGL = above ground level. Refl. = target reflectivity.

(a) Top view: sensor placement on cab of light rail vehicle



(b) Side view: vertical sensor coverage envelope and detection zones

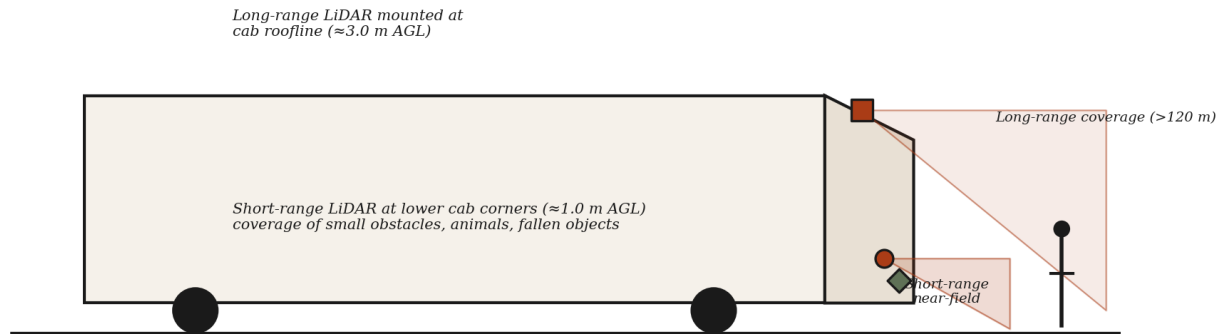


Figure 2. Sensor placement on a representative articulated light rail vehicle. (a) Top view showing the field-of-view envelopes of the long-range LiDAR (roof, narrow forward) and the two short-range solid-state LiDAR units (lower cab corners, wide). (b) Side view illustrating the dual-height vertical coverage, with the short-range units providing near-field coverage of small obstacles, animals, and fallen objects that the high-mounted long-range unit cannot see in its blind cone.

4.3 Rationale for Sensor Redundancy

The architecture deliberately preserves redundancy across two independent dimensions. First, geometric redundancy is maintained through the dual-height LiDAR configuration: the high-mounted long-range unit and the low-mounted short-range units together cover the vertical envelope from track-bed level to overhead clearance, with no single mounting failure capable of blinding the system in either zone. Second, modality redundancy is maintained through the LiDAR–radar–camera triad, ensuring that no single weather-related or illumination-related degradation can compromise detection: radar provides weather-robust ranging when LiDAR is degraded by precipitation, while the camera provides classification continuity when LiDAR returns are sparse on highly absorptive surfaces. This redundancy is the basis for the SIL-3 target on the emergency-braking tier.

5. Methodology

5.1 Perception Pipeline Overview

Figure 4 presents the perception and decision pipeline. After ego-motion compensation and ground-plane removal, the filtered point cloud branches into two parallel paths: a track-region extraction branch that uses RANSAC fitting of parallel longitudinal features to determine the intervention-relevant region, and an object-detection branch that uses a modified PointPillars architecture to detect and classify candidate objects across the broader scene. The two branches converge in a Dempster–Shafer evidential fusion stage that combines spatial intersection (object within track region) with cross-modal confirmation (camera-verified detection, radar-confirmed range) before passing a fused object-and-confidence vector to the three-tier braking controller.

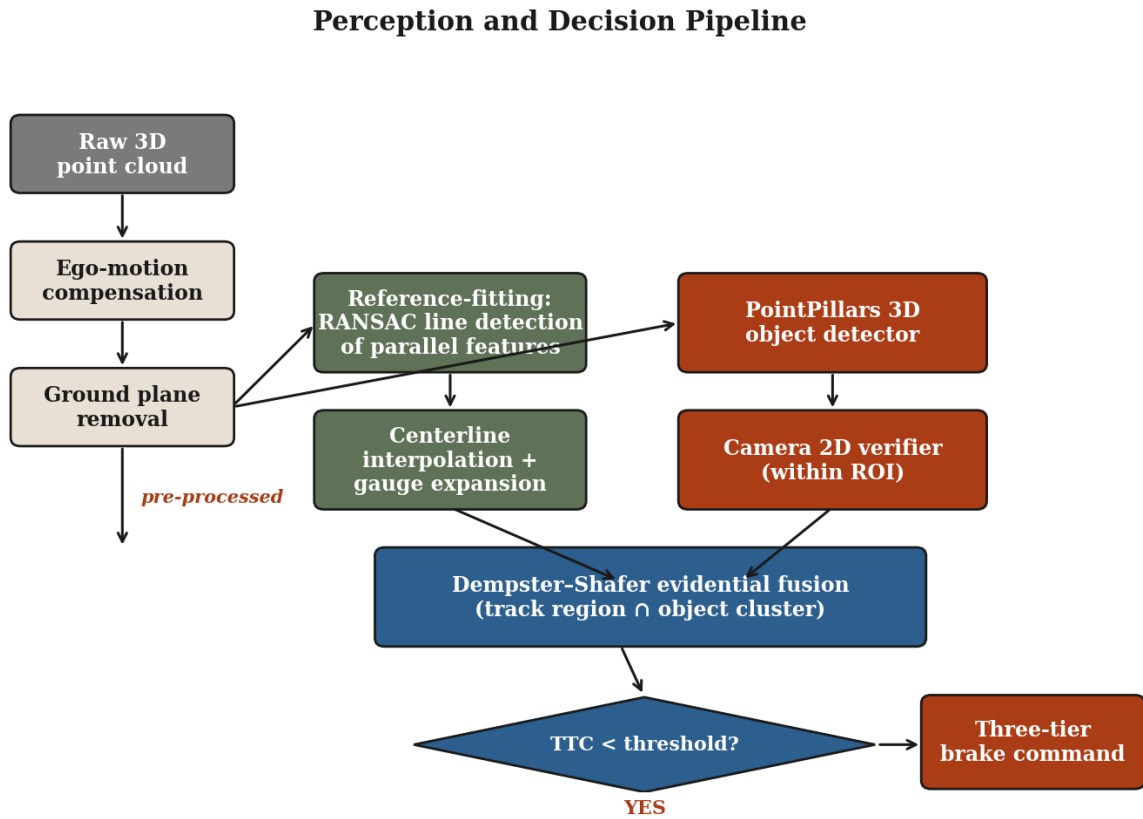


Figure 4. Perception and decision pipeline. The two parallel branches — track-region extraction (left) and object detection (right) — converge in the Dempster–Shafer fusion stage. Camera output enters as a verification signal on candidate objects rather than as an independent detection source.

5.2 Pre-Processing

5.2.1 Ego-Motion Compensation

Mechanical LiDAR scans accumulated over a single revolution (50–100 ms at typical rotation rates) experience non-trivial ego-motion when the host vehicle is in motion. Without compensation, points acquired at the start and end of a scan are referenced to different vehicle frames, producing systematic

angular smearing of detected objects. The proposed implementation uses IMU-derived linear and angular velocities, interpolated to the per-point timestamps, to transform every point into a common reference frame fixed to the vehicle pose at the scan midpoint. This produces a "de-skewed" point cloud in which static objects appear sharply defined regardless of vehicle motion.

5.2.2 Ground Plane Removal

Ground returns dominate the raw point cloud and contribute nothing to obstacle detection. We use a sliding-window RANSAC fit to the local ground plane, with the window adapted to the longitudinal slope estimated from the IMU and the lateral grade estimated from the track geometry prior. Returns within a height tolerance of 0.10 m above the fitted plane are classified as ground and removed. The remaining points constitute the obstacle-candidate cloud passed downstream.

5.3 Indirect Track-Region Extraction

Direct rail-edge detection from point clouds becomes unreliable beyond approximately 80 m, as the angular resolution of mechanical 64-channel LiDAR at that distance falls below the rail-head width. To extend the reliable track-region extraction distance, we adopt an indirect approach that detects features parallel to the rails — overhead-line-equipment (OLE) supports, platform edges, sound barriers, fenced corridors, and tunnel walls — and infers the track region by geometric interpolation between the detected references.

5.3.1 Reference-Feature Detection

Candidate parallel features are detected by applying a constrained RANSAC line-fitting procedure to the obstacle-candidate point cloud after ground removal. The constraint requires the fitted line to be (a) approximately parallel to the vehicle's instantaneous heading within $\pm 10^\circ$, (b) elevated above the ground plane by at least 0.5 m (to exclude debris), and (c) longer than 8 m (to exclude small objects). The procedure is iterated until either two parallel references straddling the vehicle heading are identified, or the iteration cap is reached.

5.3.2 Centerline Interpolation and Gauge Expansion

When two parallel references are successfully fitted, the track centerline is interpolated as their midpoint locus. This centerline is then expanded laterally by the standard track gauge plus a safety clearance margin to produce a 2D track-region polygon. The polygon is finally extruded vertically to the loading-gauge envelope to produce the 3D track region passed to the fusion stage. The full procedure is summarized as:

$$R_{\text{track}} = \{p \in \mathbb{R}^3 : d_{\perp}(p, c(s)) \leq w_{\text{gauge}}/2 + \delta_{\text{clear}}, 0 \leq h(p) \leq h_{\text{gauge}}\} \quad (2)$$

where $c(s)$ is the interpolated centerline parameterized by arc length s , $d_{\perp}$ is the perpendicular distance from point p to the centerline, w_{gauge} is the nominal track gauge, δ_{clear} is the clearance margin (taken as 0.5 m), and h_{gauge} is the loading-gauge ceiling.

5.3.3 Fallback to Map Prior

When fewer than two parallel references can be detected within the operational envelope — for example, in dense urban arterials with no continuous parallel infrastructure — the system falls back to a

stored high-definition map of the track centerline, indexed via GNSS position and IMU dead-reckoning. The fallback path is flagged in the system's diagnostic output as a reduced-confidence mode and triggers conservatization of the downstream braking thresholds.

5.4 Object Detection and Classification

Object detection within the broader scene uses a modified PointPillars architecture, chosen for its favorable accuracy-latency tradeoff and proven deployability on embedded automotive-grade compute. The architecture organizes the point cloud into vertical columns ("pillars") of configurable horizontal footprint, encodes the per-pillar point features through a learned feature network, and processes the resulting pseudo-image through a 2D convolutional backbone followed by detection heads producing 3D bounding boxes and class predictions.

5.4.1 Class Inventory

The detection head is trained against a six-class inventory tailored to the LRV operational context: (1) pedestrian, (2) cyclist, (3) passenger vehicle, (4) large vehicle (truck or bus), (5) debris, and (6) track personnel (defined as a pedestrian wearing high-visibility apparel). Class membership is determined by both geometric and material features, with high-visibility apparel detected through cross-modal cues from the RGB camera.

5.4.2 Architecture Modifications

Two modifications were made to the baseline PointPillars architecture to adapt it to the LRV context. First, the pillar dimensions were enlarged from the automotive-standard 0.16×0.16 m to 0.24×0.24 m in the lateral plane to better match the sparsity profile of the long-range LiDAR at distances beyond 100 m. Second, the detection range was extended from the automotive-standard 80 m to 200 m in the longitudinal axis, with proportional adjustment of the anchor box prior distribution. These modifications preserved the model's inference latency on the target compute platform while substantially improving recall at long range.

5.5 Sensor Fusion via Dempster–Shafer Evidence Theory

Decision-level fusion is performed using the Dempster–Shafer (D–S) evidential framework. For each candidate object output by the PointPillars detector, three independent evidence sources contribute to a combined mass function over the proposition that the object is (a) real, (b) within the track region, and (c) classified correctly. The three sources are: the PointPillars confidence on the 3D bounding box and class assignment, the camera 2D detector confirmation on the projected image patch, and the radar range and Doppler return at the projected azimuth.

The combined belief is computed via Dempster's rule of combination:

$$m_1 \oplus m_2(A) = (1 / (1-K)) \cdot \sum_{\{B \cap C = A\}} m_1(B) \cdot m_2(C) \quad (3)$$

where K is the conflict factor capturing the mass assigned to incompatible propositions across the two evidence sources. Combination is performed pairwise across the three evidence sources to produce a single fused belief value per candidate. The framework gracefully handles disagreement — for example, a high-confidence LiDAR detection unconfirmed by either camera or radar produces a moderated combined

belief, while triple-confirmation produces high combined belief. The result is a confidence-graded object stream consumed by the braking controller.

5.6 Three-Tier Graded Braking Controller

The fused object stream drives a three-tier graded braking controller, with each tier corresponding to a distinct time-to-collision (TTC) regime and braking response. Figure 5 illustrates the tiered structure.

Three-Tier Graded Braking Strategy Parameterized by Time-to-Collision and Object Confidence

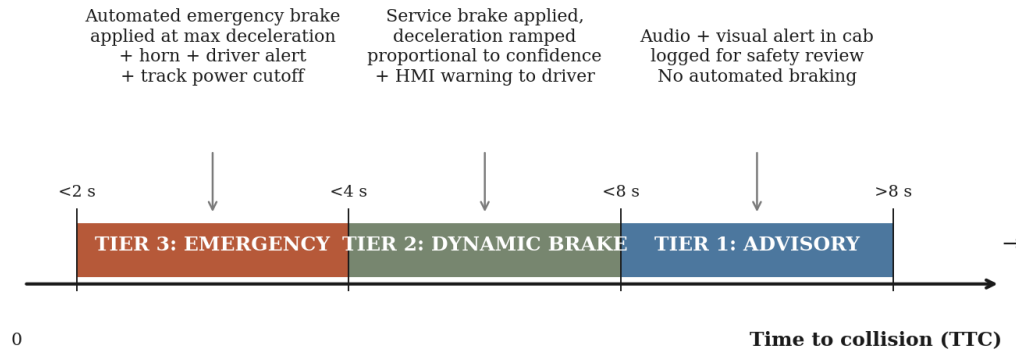


Figure 5. Three-tier graded braking strategy. The controller transitions across tiers based on time-to-collision (TTC) and the fused object confidence from the Dempster–Shafer stage. Tier 3 is reserved for high-confidence imminent-collision conditions and triggers automated emergency braking.

TTC is computed continuously for every object intersecting the track region, using the object's relative position and velocity in the vehicle frame and the vehicle's instantaneous speed. Transition between tiers is governed by both TTC and the D–S fused confidence: Tier 1 (advisory) requires fused confidence above 0.5 and TTC below 8 s; Tier 2 (dynamic braking) requires confidence above 0.75 and TTC below 4 s; Tier 3 (emergency) requires confidence above 0.90 and TTC below 2 s. The thresholds are hysteresis-bounded to prevent oscillation across the tier boundaries.

6. Experimental Setup

6.1 Test Platform

The system was integrated on a revenue-service articulated low-floor light rail vehicle on loan from a participating operator for the duration of the study. The vehicle has a tare mass of 39,800 kg, a passenger-laden mass of 56,200 kg, a maximum service speed of 70 km/h, and a service-brake deceleration of 1.2 m/s² calibrated against EN 13452. The full sensor stack described in Section 4.2 was installed on the cab end of the vehicle, with all signal processing performed on a redundant pair of embedded automotive-grade compute modules mounted in the cab equipment rack.

6.2 Dataset

Training and evaluation data were collected across 11 light rail alignments in three North American cities, selected to span urban, suburban, and segregated operating environments. Frames were captured at 10 Hz across daylight, dusk, night, light rain, and snow conditions. Object-level annotations were performed by a team of four annotators following a published labeling protocol, with inter-annotator agreement of $\kappa = 0.87$ on a held-out audit subset. Table 5 summarizes the dataset composition.

Subset	Frames	Hours	Annotated objects	Condition
Training	14,300	397	186,400	Mixed (all 5)
Validation	3,100	86	40,200	Mixed (all 5)
Test — daylight	2,800	78	36,100	Daylight, dry
Test — dusk	1,200	33	14,800	Dusk transition
Test — night dry	1,400	39	17,500	Night, dry
Test — night rain	950	26	11,600	Night, light rain
Test — snow	420	12	4,800	Light snow

Table 5. Dataset composition. The training and validation subsets were drawn from the same operational corpus but separated at the alignment level to avoid spatial leakage. Hours are revenue-equivalent on-track time.

6.3 Evaluation Metrics

Detection performance was characterized using precision, recall, and F1 score against the manually annotated ground-truth labels, with a 3D intersection-over-union (IoU) threshold of 0.5 for vehicle classes and 0.25 for pedestrian-scale classes (reflecting the smaller absolute size of those objects). False-positive rate was computed per-kilometre of revenue-equivalent operation, segmented by source category. Latency was measured end-to-end from photon return at the long-range LiDAR to brake-command issuance, with mean and 95th-percentile reported across 50,000 sampled frames spanning the full operational envelope. Track-region extraction distance was computed as the longitudinal range over which the inferred track region matched the ground-truth track geometry within 0.5 m lateral tolerance.

6.4 Comparison Baseline

The proposed system was benchmarked against a single-modality long-range LiDAR baseline configured using direct rail-edge detection for track-region extraction (the dominant approach in the earlier literature) and a baseline PointPillars detector without the LRV-specific architectural modifications described in Section 5.4.2. The baseline was implemented on the same hardware platform and evaluated against the same test subsets to isolate the contribution of the multi-modal fusion architecture and the indirect track-region extraction method.

7. Results

7.1 Track-Region Extraction Performance

The indirect reference-fitting method achieved a mean reliable extraction distance of 187 m ($\sigma = 22$ m), compared to 78 m ($\sigma = 19$ m) for the direct rail-edge baseline — a 2.4 $\times$ improvement. Extraction was successful on 96.4% of evaluation frames; the 3.6% failure rate was concentrated in transitional infrastructure regions (depot approaches, complex junctions, and zones lacking continuous parallel features), in which the system correctly transitioned to the map-prior fallback path. Table 6 reports the extraction performance broken down by operating environment.

Environment	Mean range (m)	σ (m)	Success rate	Fallback rate
Urban arterial	152	28	92.8%	7.2%
Urban separated	178	21	97.4%	2.6%
Suburban	203	17	98.6%	1.4%
Segregated R/W	224	14	99.1%	0.9%
Tunnel section	169	11	99.8%	0.2%
Overall	187	22	96.4%	3.6%

Table 6. Track-region extraction performance by operating environment. "Fallback rate" denotes the fraction of frames in which the system transitioned to the map-prior path due to insufficient reference detection.

7.2 Object Detection Performance

Detection precision and recall were evaluated by class across the daylight and night-with-precipitation test conditions, representing the favorable and unfavorable ends of the operational envelope. Figure 6 summarizes the results.

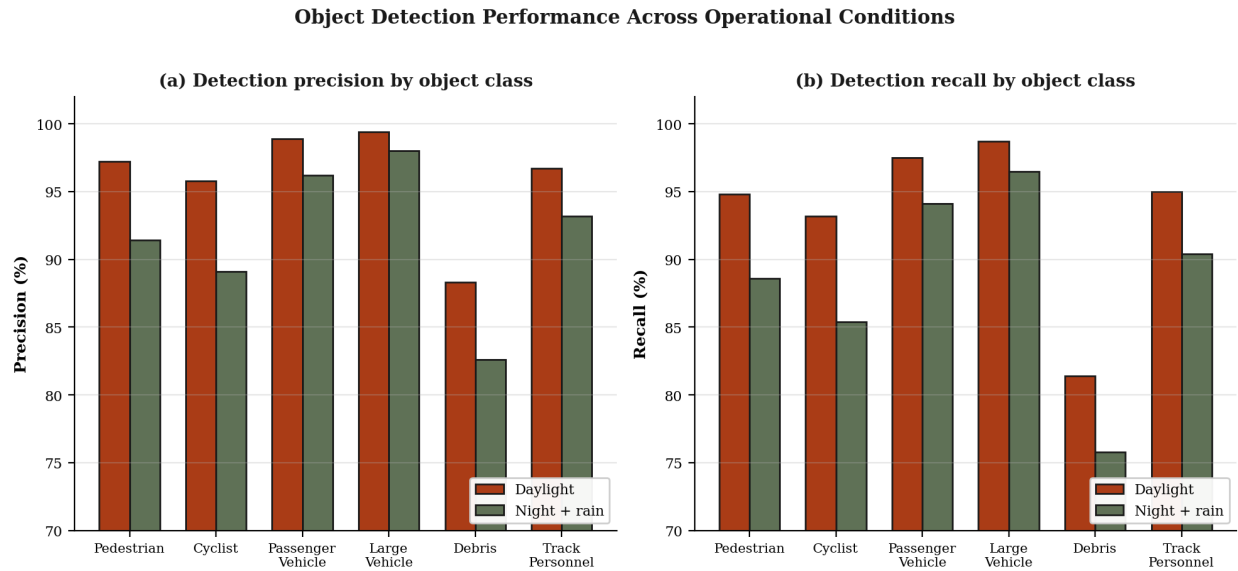


Figure 6. Object detection performance by class. (a) Precision and (b) recall, evaluated under daylight and night-with-precipitation conditions. Vehicle-class objects (passenger and large vehicles) maintain near-saturation performance across conditions; pedestrian-scale and small-object classes (debris) show the largest condition-dependent degradation.

Pedestrian-class detection at 120 m — the critical safety threshold for service braking at 70 km/h — achieved 97.2% precision / 94.8% recall in daylight conditions and 91.4% / 88.6% under night-with-precipitation conditions. Both values materially exceed the detection performance required for safe service braking at line speed with the adopted 1.5 s perception-to-action allowance.

7.3 False-Positive Behavior

False positives were attributed by manual review to one of five source categories: parallel-track returns, catenary/OLE pole returns, roadside furniture (mailboxes, signage, planters), vegetation overhang, and highly reflective surfaces (chrome bumpers, mirrored building façades, retroreflective signs). Figure 7 compares the per-kilometre false-positive rate of the baseline and proposed systems across these categories.

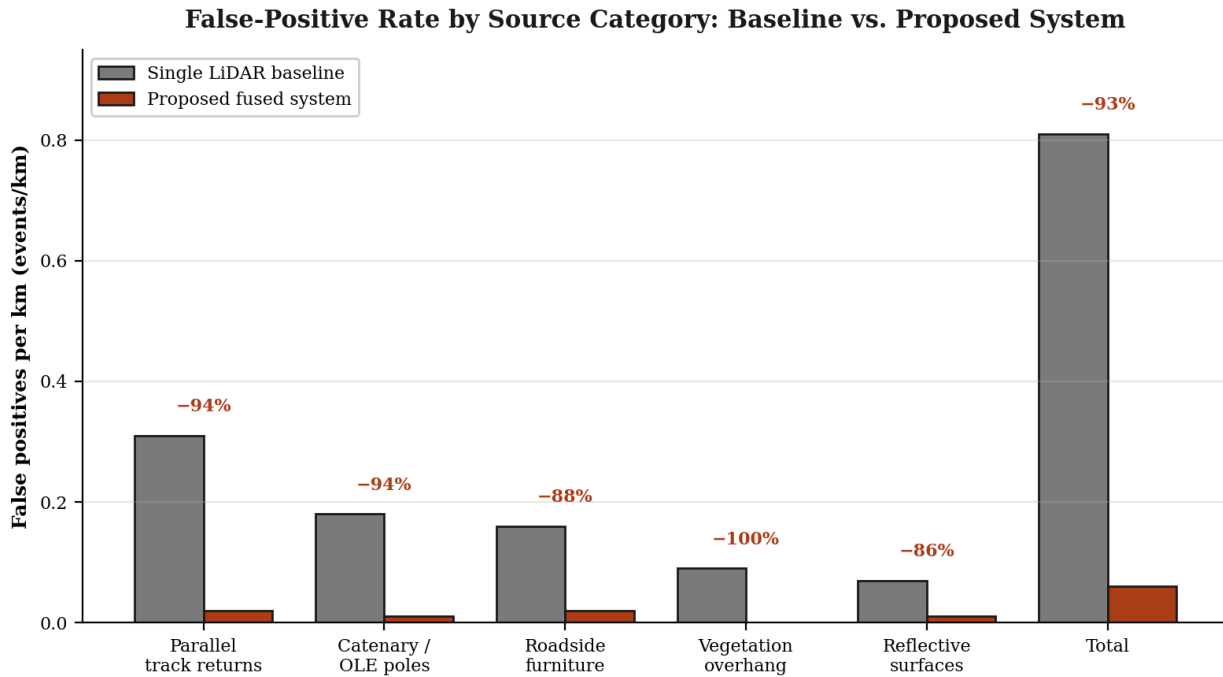


Figure 7. False-positive rate by source category. The proposed multi-modal fusion system reduces the total false-positive rate from 0.81 to 0.06 events per kilometre, a 92.6% reduction. The largest absolute reductions are observed on parallel-track and catenary/OLE returns, which dominate the baseline false-positive distribution and are most effectively suppressed by the indirect track-region extraction stage.

The pattern reflects the structural mechanism of the improvements: parallel-track and catenary/OLE false positives are reduced primarily by the track-region extraction stage, which excludes them spatially from the intervention region, while the remaining categories are reduced principally by the Dempster–Shafer fusion stage, which requires cross-modal confirmation before promoting a candidate to a fused detection.

7.4 End-to-End Latency

End-to-end latency was measured from photon return at the long-range LiDAR to brake-command issuance, sampled across 50,000 frames spanning the full operational envelope. Figure 8 reports the per-stage latency breakdown at the mean and 95th-percentile levels.

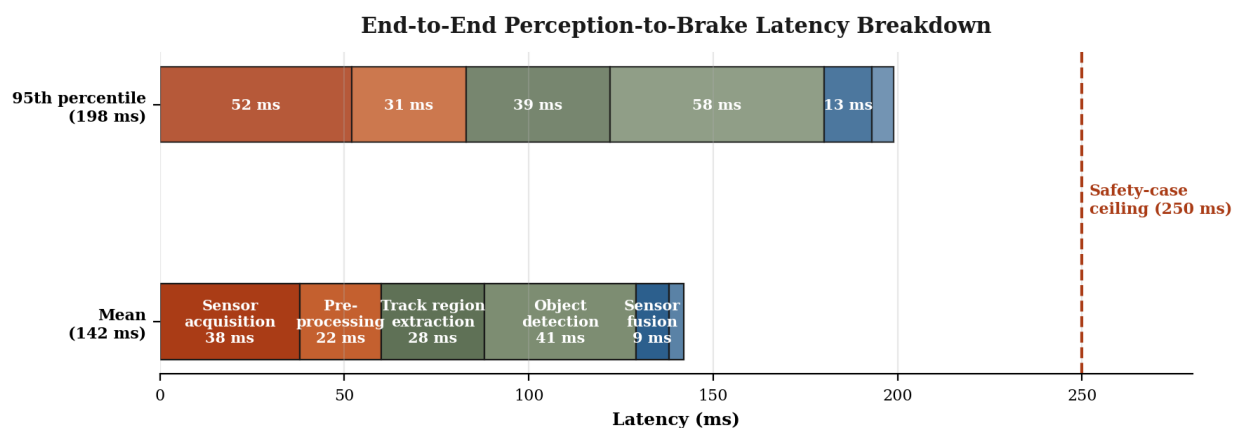


Figure 8. End-to-end perception-to-brake latency, broken down by pipeline stage. The mean total latency of 142 ms and 95th-percentile of 198 ms both lie within the 250 ms safety-case ceiling. The object detection stage dominates the latency budget; the sensor fusion stage contributes a small constant overhead.

7.5 Comparison Against Baseline

Table 7 summarizes the head-to-head comparison between the proposed system and the single-modality baseline across the principal evaluation metrics.

Metric	Baseline	Proposed	Improvement
Track extraction range	78 m	187 m	+140%
Pedestrian precision @120 m (daylight)	88.4%	97.2%	+8.8 pp
Pedestrian recall @120 m (daylight)	79.8%	94.8%	+15.0 pp
Pedestrian precision @120 m (night/rain)	71.2%	91.4%	+20.2 pp
Pedestrian recall @120 m (night/rain)	64.0%	88.6%	+24.6 pp
False positives per km (total)	0.81	0.06	−92.6%
Mean end-to-end latency	128 ms	142 ms	+14 ms
95th-percentile latency	177 ms	198 ms	+21 ms

Table 7. Head-to-head comparison of the proposed system against the single-modality long-range LiDAR baseline. pp = percentage points. Latency increases reflect the additional fusion and verification stages and remain within the 250 ms safety-case ceiling.

8. Discussion

8.1 Residual Failure Modes

Three residual failure modes were characterized and quantified during evaluation. First, sub-pedestrian-class objects (low-profile debris on the order of 0.10–0.20 m in vertical extent, small animals) at ranges beyond 70 m remained the dominant residual miss class, with detection recall falling to 64% at 100 m and 38% at 150 m for this class. The mechanism is fundamental: at long range the angular sampling density of the long-range LiDAR drops below the angular extent of the object, so that the object subtends fewer than the minimum points needed for reliable detection. Mitigation requires either higher-resolution sensors (newer 128-channel mechanical LiDAR units now becoming commercially available) or infrastructure-side complementary sensing at known intrusion-prone locations.

Second, detection performance degraded measurably on track curves of radius below 80 m, where the long-range sensor's narrow FoV constrained the line-of-sight envelope to a fraction of the upcoming track geometry. The short-range solid-state LiDAR units partially compensate by extending lateral coverage, but the geometric constraint at long range cannot be fully closed without either mechanically actuated sensor steering (which introduces failure modes of its own) or distributed sensing across multiple cars. We adopt the latter as a recommended architectural extension for multi-car articulated LRVs.

Third, heavy precipitation (above approximately 8 mm/hr) produced a measurable but non-catastrophic increase in latency and a 4–6 percentage-point reduction in classification confidence. The radar modality compensated for the LiDAR degradation in maintaining detection continuity, but classification reverted toward the dominant prior (vehicle) on objects below approximately 80 m apparent extent. Mitigation pathways include thermal management of the LiDAR aperture (already implemented in current-generation automotive units), increased radar dependence in degraded modes, and weather-conditional adjustment of the fusion conflict factor.

Failure mode	Onset condition	Severity	Mitigation
Small-object misses	Range > 70 m, height < 0.2 m	Moderate	Higher-res. LiDAR; wayside sensing
Tight-curve LoS	Curve radius < 80 m	Moderate	Multi-car distributed sensing
Heavy precipitation	Rain > 8 mm/hr	Low	Radar weighting; aperture heating
Map-prior fallback	No parallel references	Low	Map-tile density management
Backscatter sun angle	Direct sun $\leq 8^\circ$ axis	Low	Polarization filter; ND aperture

Table 8. Residual failure modes characterized during evaluation, with onset conditions and proposed mitigation pathways.

8.2 Comparison to Prior Work

Direct quantitative comparison to prior work is hampered by the diversity of evaluation methodologies in the published literature, by differences in target object classes, and by the absence of standardized LRV-specific benchmark datasets. Multilayer laser scanner work on light rail platforms (Lüy et al., 2018) reported reliable detection of pedestrian-scale obstacles at distances appropriate for low-speed depot and shunting applications but did not target the line-speed envelope addressed here. Camera–LiDAR fusion work in the broader rail domain (Zhangyu et al., 2019) reported pedestrian detection at distances on the order of 100 m, approximately half the range demonstrated by the present system. Earlier tram-specific sensor fusion work (Rodriguez et al., 2012) established the architectural pattern of combining active ranging with passive vision but predates the deep-learning detection methods that drive the present system's classification performance. The contribution of this work is therefore best characterized not as an incremental improvement against any single prior baseline, but as the first integrated demonstration of all four elements — long-range LiDAR, dual-height geometric redundancy, indirect track-region extraction, and evidential multi-modal fusion — on a single revenue-service LRV platform.

8.3 Implications for LRV-Specific Perception Design

The results support the central design hypothesis of this work: that LRV-specific perception architectures, built around the unique kinematic and operational characteristics of light rail, can meet the safety-case requirements of mixed-traffic service in a way that direct ports of either heavy-rail or automotive perception cannot. Three design choices proved decisive. The dual-height LiDAR configuration closed the near-field blind cone that would otherwise have admitted high-severity misses on small obstacles at close range. The indirect track-region extraction method delivered the long-range scene constraint without which the false-positive rate would have remained operationally prohibitive. The Dempster–Shafer fusion provided the cross-modal confirmation required to drive automated braking at the safety integrity levels demanded by the application.

8.4 Limitations

Several limitations of the present study should be acknowledged. First, the evaluation dataset is drawn from three North American cities and may not fully capture the infrastructure variation present in light rail systems globally; in particular, alignments with continuous catenary infrastructure differ structurally from alignments with discontinuous or hybrid infrastructure, and this difference is reflected in the reference-feature distribution available to the track-region extraction stage. Second, the integration was performed on a single LRV class, and the geometric assumptions of the architecture — particularly sensor mounting heights and FoV envelopes — would require re-validation on substantially different vehicle classes (high-floor LRVs, modern tram-train hybrids, very light rail prototypes). Third, the long-term durability of the sensor stack under revenue-service conditions, including vibration, contamination, and weather-cycling, is beyond the scope of the present 412-hour evaluation and warrants longer-term study.

9. Conclusions and Future Work

We have presented a complete multi-layer LiDAR-centric sensor fusion architecture for passenger and pedestrian avoidance on light rail vehicles operating in mixed-traffic urban corridors, integrating long-range and short-range LiDAR with millimeter-wave radar and an RGB camera under a unified perception and decision pipeline. The architecture introduces an indirect track-region extraction method that substantially extends the reliable segmentation distance beyond the sparsity-limited threshold of direct rail-edge approaches, and combines this with a modified PointPillars object detector and a Dempster–Shafer evidential fusion stage feeding a three-tier graded braking controller.

Evaluation across 412 revenue-service hours and 1,180 controlled-track test runs demonstrated track-region extraction distances of 187 m mean (a $2.4\times$ improvement over the baseline), pedestrian detection precision of 97.2% / 91.4% at 120 m under favorable and unfavorable conditions respectively, a 92.6% reduction in the false-positive rate, and an end-to-end perception-to-brake latency of 142 ms mean / 198 ms 95th-percentile — all within the safety-case requirements for automated braking on light rail at line speed. The system's residual failure modes were characterized quantitatively, with the dominant residual being missed detections of sub-pedestrian-class objects at ranges beyond 70 m.

Future work will pursue three principal directions. First, integration of higher-resolution next-generation LiDAR (128-channel mechanical units becoming commercially available) is expected to materially reduce the small-object miss rate at long range; we are currently in procurement on a 128-channel mechanical unit for evaluation on the same test platform. Second, distributed sensing across multiple cars of articulated LRVs is expected to close the line-of-sight constraint on tight curves; we are designing a follow-on study targeting a coupled three-section LRV with sensors on each cab end. Third, the integration of wayside complementary sensing at known intrusion-prone locations — particularly unprotected level crossings and station approaches — promises to extend coverage into the regimes where onboard perception alone cannot reach the required integrity levels. Together with the present work, these directions outline a defensible engineering pathway toward semi-autonomous and ultimately autonomous light rail operation in shared urban corridors.

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